

Psuedocodes

Surface Blasting Design Psuedocode

Start

DISPLAY "BlastSmart Surface Blasting Calculator"

INPUT BlastHoleDiameter

INPUT AveragelnholeDensity

INPUT SpacingBurdenRatio

INPUT BenchHeight

Burden = $K \times \text{Blasthole Diameter (BD)}$

StemmingLength = $(\text{Blasthole Diameter}/1000) \times 30$

ActualPowderFactor = $(\text{Average Inhole Density} \times (\text{Bench Height} - \text{Stemming Length})) / (\text{Burden} \times \text{S/B ratio} \times \text{Bench Height})$

DISPLAY "Burden =", Burden

DISPLAY "Stemming Length =", StemmingLength

DISPLAY "Actual Powder Factor =", ActualPowderFactor

End

Fragmentation Calculator

START

DISPLAY "BlastSmart Fragmentation Calculator"

INPUT AveragelnholeDensity

INPUT BenchHeight

INPUT RelativeEffectiveEnergy

INPUT StemmingLength

MeanFragmentSize = KuzRamFormula

DISPLAY "Mean Fragment Size =", MeanFragmentSize

End